

Integrating Computing in Physics Courses

The Landscape of Computing in Physics Education

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Abstract

Computing is a core part of physics, and the use of computational tools in physics courses is becoming increasingly common. The landscape of computing in physics courses is complex and diverse, with a wide range of tools, languages, and pedagogical approaches being used. In this edited volume, we collect a series of stories from educators around the world who have integrated computing into their physics courses. These stories highlight the diversity of approaches to computing in physics education, and the challenges and opportunities that come with them. We hope that this volume will serve as a resource for physics educators who are interested in integrating computing into their courses, and as a starting point for discussions about the future of computing in physics education.

Keywords computing, physics, education, institutional change, curriculum, pedagogy, professional development

1. INTRODUCTION

Computing has long been central to the enterprise of physics. We run experiments and collect data using computers; we build models and develop simulations with a variety of programming languages and tools; and we design studies and validate experimental results through diverse computational methods. At the largest experimental scales, computing is essential for controlling massive systems like the Large Hadron Collider and the Laser Interferometer Gravitational-Wave Observatory—two experiments that have produced some of the most exciting physics in recent years (Aad et al., 2012; Abbott et al., 2016). Enabling these discoveries required advanced computational modeling and analysis techniques, some of which were recognized with the 2024 Nobel Prize in Physics (Fattaruso, 2024). Large-scale modeling and simulation are pervasive in physics research and are widely used to understand complex fluids (Duraismy et al., 2019; Font, 2008), astrophysical phenomena (Mezzacappa, 2020; Somerville & Davé, 2015), and many-body systems (Lee, 2009; Nandkishore & Huse, 2015). Computing is equally critical in table-top physics, providing experimental control, data collection, and analysis and modeling tools that enable the daily research of physicists. Without computing, the

advancement of physics would have been much slower, and many of our most important discoveries might have been missed.

While computing has revolutionized physics research, our physics courses and curricula have not always reflected the importance and utility of computing (Chonacky & Winch, 2008). As personal computers became more affordable and accessible, and as more programming languages were developed, some educators began working to integrate computing into physics courses. Early examples were limited to physics majors (MacDonald et al., 1988), to specific programs with particular expertise or resources (Niedderer et al., 1991; Schecker, 1994), or to novel educational efforts (Abelson & DiSessa, 1986; Sherin et al., 1993). These early efforts were not designed to be adaptable to other contexts, yet they provided important curricular and pedagogical lessons that eventually informed more comprehensive initiatives adopted by additional institutions and programs.

The increasing affordability of computing devices, the growth of networked computers, and the integration of technology into university and K-12 programs further enabled these efforts. Coinciding with these advances were efforts to expand the role of computer science in K-12 and teacher preparation programs (National Research Council, 2012; Olson & Riordan, 2012; Vogel et al., 2017). National calls to integrate computing into K-16 classrooms were issued (AAPT Undergraduate Curriculum Task Force, 2016; Behringer & Engelhardt, 2017; Caballero et al., 2020; Joint Task Force on Undergraduate Physics Programs, 2016), identifying potential learning goals for different levels of physics instruction and anticipating forthcoming implementation challenges. Caballero & Odden (2024) traces these efforts over time and provides a thematic grouping of this work. Physics and other science programs began more strategic efforts to integrate computation.

The physics education community recognizes that our students are entering a scientific workforce where an understanding of computing is essential. In response to these and other calls, educators, curriculum developers, and education researchers have worked in earnest to integrate computing into physics learning spaces. This work has included design efforts (Burke & Atherton, 2017; Caballero et al., 2014; Caballero & Pollock, 2014; Chabay & Sherwood, 2008; Cook, 2008; Irving et al., 2017; McIntyre et al., 2008; Odden & Caballero, 2019) and investigations into student understanding and engagement (Caballero et al., 2012; Kohlmyer, 2005; Lunk & Beichner, 2016; Obsniuk et al., 2015; Sand et al., 2018; Weatherford, 2011; Weintrop et al., 2016), as well as broader studies of the mechanics and dynamics of teaching and learning in computing-enabled physics classrooms (Bott et al., 2019; Byun et al., 2023; Christensen et al., 2022; Leary et al., 2018; Obsniuk, 2020; Pawlak et al., 2020; Shah et al., 2020; 2021; Silvia et al., 2023; Stroupe et al., 2022; Vieyra et al., 2024; Weller et al., 2019; Young et al., 2019). These recent efforts have demonstrated that computing can be integrated into physics education at various stages of learning and that such integrations can enhance both conceptual understanding and problem-solving skills. While these efforts are impressive and necessary, they do not reach all programs, instructors, or students. Nor is our understanding of how computing can best support

student learning, engagement, and growth by any means complete (Caballero & Odden, 2024).

These advances would not have been possible without the community of educators who have worked together to share ideas, approaches, pitfalls, and successes. Physics educators shared early lessons from their classrooms and provided the software and activities they delivered to students in a free and open way (Christian et al., 2011; Esquembre, 2004; Scherer et al., 2000). The drive to ensure these materials were freely available and open-source allowed the physics education community to access them readily. Yet, the use of computing in physics classrooms remained limited to those with prior experience or interest in computing (Caballero & Merner, 2018; Chonacky & Winch, 2008). Without a more strategic effort to connect physics educators and build a community of support that honored these early lessons, the growth of computing in physics classrooms would have remained limited.

The Partnership for the Integration of Computation in Undergraduate Physics (PICUP) was founded to address these challenges (Caballero et al., 2019; Partnership for Integration of Computation into Undergraduate Physics, 2025). With a successful series of National Science Foundation (NSF) grants, PICUP designed impactful professional development opportunities and fostered a community of university physics educators dedicated to integrating computing into their courses (Caballero et al., 2022). Efforts to grow other networks of high school (as well as K-8) educators teaching computing in physics have also been successful (Apple et al., 2021; Orban & Teeling-Smith, 2020; Shah et al., 2020; Stroupe et al., 2025; Vieyra et al., 2024; Weintrop et al., 2016; Weller et al., 2022). There is significant overlap among these efforts, with researchers and educators often engaging in multiple initiatives simultaneously.

This chapter introduces the problem statement for the book:

How do we integrate computing into physics courses in sustainable ways that lead to deep learning and robust understanding?

There are numerous ways to answer this question. Different perspectives are valuable as we collectively develop a deeper understanding of what happens when students learn computing in their physics courses. Organizing and making sense of these perspectives can be challenging, as there are overlapping and intersecting issues and concerns. This is simply the nature of institutional change—it's a complicated affair.

One way to organize these ideas is to consider the particular *Frame of Context* on which we intend to focus (Cole, 1998; Finkelstein, 2005). These frames highlight different aspects of our integration efforts and illustrate how those aspects are connected. For computational physics education, I have used the nested frames shown in [Figure 1](#) to help educators identify these different frames and understand how they can intersect. For example, the university or college may have certain resources and expectations that shape how the physics department operates. The

relevant courses might appear in specific parts of the physics curriculum, grounded in particular goals and using certain resources and learning spaces. The class meeting is further influenced by these considerations, and its particular affordances and constraints make some activities more feasible than others. The specific tasks that students perform may themselves have conceptual, procedural, or other goals that need to be aligned with the broader frames. These nested frames provide a way of thinking about which elements to consider as we integrate computing into physics courses. This model also acknowledges the interacting nature of each frame, as each is nested within the one that subsumes it.

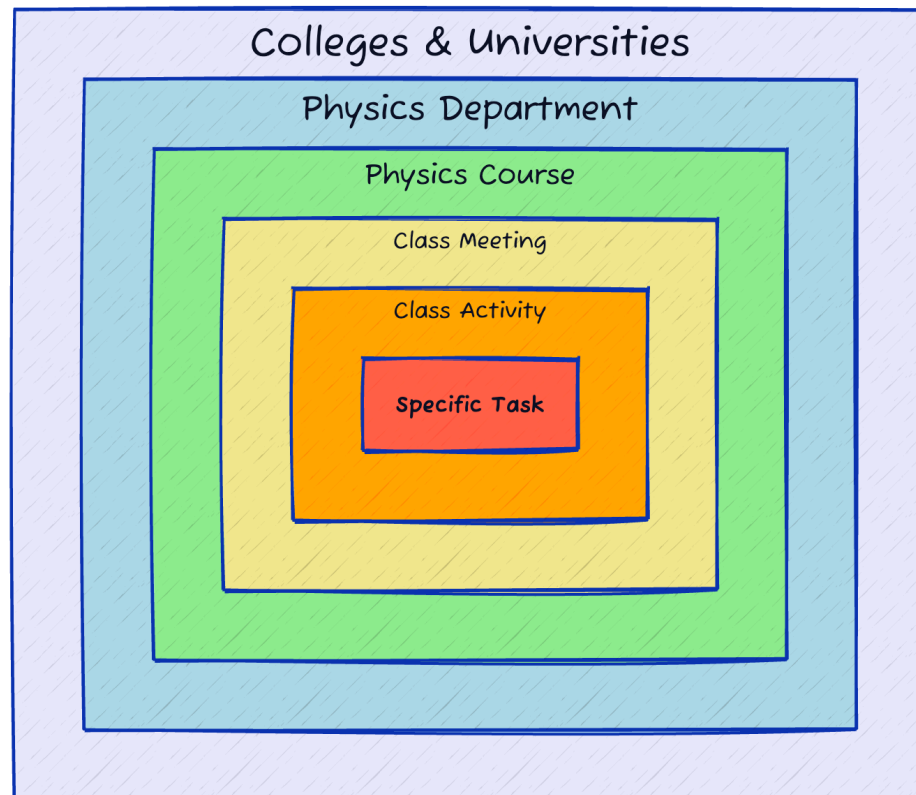


Figure 1: Frames of Context for Computational Physics Education, inspired by Finkelstein (2005). The nested nature of these frames highlights the focus of our efforts to integrate computing into physics courses, while also acknowledging the intersecting nature of each element. Sustainable integration of computing in physics courses requires addressing each of these frames.

The particular frame we take is further complicated by the perspectives and backgrounds we bring to the question: *How do we integrate computing into physics courses in sustainable ways that lead to deep learning and robust understanding?* Consider your role and expertise in the teaching and learning of physics. Are you a physics educator, a research physicist, an education researcher, a curriculum developer, or a professional development specialist? Or do you take on these roles at different times? How confident do you feel in each of these roles? Ideas and strategies developed from each of these perspectives are important for building a

comprehensive effort that addresses the need for computing in physics coursework. In this book, I have invited authors to speak about different frames of and from different perspectives on integrating computing into physics courses, in order to respect that diversity.

2. THE STATE OF COMPUTING IN PHYSICS EDUCATION

While each chapter introduces its own perspective on the state of computing in physics education, I offer an overview here, referencing some of the major milestones and publications that have shaped the field. A more detailed history appears in Odden & Caballero (2023).

We can organize the past few decades of efforts to integrate computing into physics courses into three strands:

1. *Modeling and Representations*: rooted in the tradition of physics that emphasizes constructing mathematical models of physical phenomena.
2. *Scientific Practices*: developed from work in the learning sciences on apprenticeship and enculturation.
3. *Computational Thinking*: grounded in computer science education and focused on problem-solving.

The Modeling and Representations strand takes the epistemic perspective that physics is a process of generating, developing, refining, and evaluating models of physical systems. The core idea is that the work of physics involves making quantitative models of the physical world. This view aligns with advocates for modeling as a curricular touchstone (Brewer, 2008; Halloun & Hestenes, 1985; Hestenes, 1987). Here, computing is another tool that enables modeling and the production of representations of systems. Animations, graphs, and other visualizations provide information that students can use to evaluate the quality and performance of their computational models. Examples from the Modeling and Representation strand reported in Odden & Caballero (2023) illustrate key findings:

- Students with no programming experience can learn to make computational models of physical systems (Aiken et al., 2012; Buffler et al., 2008; Caballero et al., 2012; 2014; Caballero & Pollock, 2014; Chabay & Sherwood, 2008; Irving et al., 2017; Lunk & Beichner, 2016).
- A modeling perspective can introduce students with different backgrounds in physics and mathematics to computing (Caballero et al., 2012; Svensson et al., 2020).
- Computing can be used in inquiry-based instruction and just-in-time teaching (Hutchins et al., 2020; Irving et al., 2017).
- Students can make meaning of their models and use them for sensemaking (Aiken et al., 2012; Buffler et al., 2008; Orban & Teeling-Smith, 2020; Sand et al., 2018).

The Scientific Practices strand takes the epistemic perspective that physics consists of a set of skills, ideas, and habits of mind, as well as a social community and an identity that one can adopt. These ideas stem from the tradition of situated cognition and communities of practice (Lave & Wenger, 1991; Wenger, 1999). From this view, computing is a set of practices, ideas, and ways of thinking that the physics community uses, and students should learn these through authentic social experiences in physics. While there has been less research on the impact of this perspective, there are numerous approaches that leverage this design in both introductory and upper-level courses successfully (Burke & Atherton, 2017; Irving et al., 2017; 2020; Odden & Burk, 2020; Odden & Malthe-Sørensen, 2020).

Computational Thinking (CT) is epistemically rooted in algorithmic thinking and computer science education. While the concept has a substantial history, its recent growth stems from Wing (2006), where Jeanette Wing outlined initial constructs underlying computational thinking. In Wing (2011), computational thinking is defined as:

the thought processes involved in formulating problems and their solutions so that the solutions are represented in a form that can be effectively carried out by an information-processing agent.

Here, the focus is on problem-solving, specifically on developing algorithmic solutions that a computer (i.e., an information-processing agent) can execute. The challenge with implementing this perspective is that it is contentious, and deciding what counts as CT is often unclear. To facilitate more productive discussion, Weintrop et al. (2016) developed definitions of different aspects of computational thinking in STEM classes, and Orban & Teeling-Smith (2020) offered a theoretical perspective on where and how computational thinking can manifest in physics courses.

Each of these strands has contributed to the development of curricular materials, professional development opportunities, and research studies that have advanced our understanding of how computing can be integrated into physics courses. The work in each strand has been important for the growth of computing in physics education, but they are not mutually exclusive. In fact, many efforts have drawn on multiple strands to develop a more comprehensive approach to integrating computing into physics courses (Odden & Caballero, 2023).

Most recently, we have seen how generative artificial intelligence (GenAI), enabled by commercial large language models (LLMs) such as ChatGPT and Google Gemini, can generate many of the programs, scripts, and codes that we might have previously asked students to write from scratch. The role these tools will play in the future of physics education is unclear, but there are ongoing efforts to investigate how GenAI and LLMs can be used in physics education (Avila et al., 2024; Kahaleh & Lopez, 2025; Mok et al., 2025), including in research (Ding et al., 2023; El-Adawy et al., 2024; Fussell et al., 2025; Polverini & Gregorcic, 2024; Wan & Chen, 2024). Others have

explored their accuracy and limitations in answering physics questions (Chen & Wan, 2025; Kortemeyer & Nöhl, 2025). While this work is promising and should continue, specific recommendations for using LLMs in physics classrooms or efforts enabled by GenAI are not a part of this book.

3. DIVERSE IMPLEMENTATIONS PRODUCE DIVERSE OUTCOMES

When initially approached about this book, editors from the Institute of Physics asked that I craft an implementation guide for computing in undergraduate physics education. I wrestled with the idea because speaking on behalf of a very diverse community—one that has developed a wide variety of approaches—seemed overwhelming at best. More importantly, a book authored solely by me would represent an authoritative contribution to the physics education community, which felt at odds with how these efforts have actually unfolded.

The growth and sustainability of integrating computing into undergraduate physics courses have been driven by the contributions of many over decades. This community has developed with a deep and abiding respect for the diversity of institutional factors, student resources, and educator perspectives. Together with my collaborators, we have authored an edited volume with chapters that represent the diversity of our community.

Each institution and program is different; students have different resources; and individual educators may have varying goals. Each of these elements is locally situated and important to consider as you develop a plan to integrate computing into physics coursework. Chapter authors have provided comprehensive and honest stories about their local conditions and factors. While no single effort in this book will match exactly with others, the diversity of these stories should provide confidence that many potential integrations are possible. How these efforts unfolded and what was needed to offer these courses is important; each story provides insight into the expertise and resources that authors drew on.

The diversity of our contributors demonstrates that we can frame computing in physics education in different ways depending on local factors. Research into institutional change has shown the importance of considering and honoring those local factors (Dancy et al., 2019; Henderson & Dancy, 2007)—a result repeated in efforts to integrate computing in physics (Leary et al., 2018). These factors cannot be overstated: physics students are not a monolith, and neither are physics educators. The resources, needs, goals, and constraints of each program can vary widely. Successful and sustainable integrations respect those local institutional factors. For example, PICUP's professional development efforts are rooted in the idea that lowering barriers for educators requires respecting institutional diversity (Caballero et al., 2019; 2022). Physics educators know their programs, classrooms, and students well, and drawing on that expertise has been highly effective.

The diversity of this community is welcome because it offers the opportunity for many students to engage productively with computing. It also ensures that the

physics education community has a robust discussion of the goals and values of integrating computing in physics education. As computing continues to enable scientific and engineering discoveries, the physics education community must continue to develop new courses, advance novel curricula, and deepen the community that integrates computational learning. Advancing physics education with computing is critical for the future of physics.

Once you finish this book, I hope we have effectively communicated that: (1) integrating computing into physics courses requires a local, dedicated effort; (2) there are many ways to integrate computing into any course or program; and (3) you can draw on the expertise and resources that already exist.

4. WHAT IS IN THIS BOOK?

In its chapters, this book presents a series of vignettes from physicists, physics educators, and physics education researchers who have worked to integrate computing into a variety of (mostly) undergraduate physics settings. The book is organized to guide the reader from foundational instructional tools, through course-level implementations, to program-wide curricular redesigns, and finally to broader theoretical and innovative perspectives. The chapters are sequenced to reflect this progression, with each chapter building on the themes and lessons of the previous one and setting the stage for what follows.

Chapter 1: M.D. Caballero; *Integrating Computing in Physics Courses*. This chapter frames the volume, providing an overview of the history and current state of computing in physics education. It outlines the book's focus and organizational structure, setting the stage for the practical tools and approaches that follow.

Chapter 2: B. Lunk and S. Weatherford; *Student Engagement with Computational Models and the Development of the “Minimally Working Program” Instructional Tool*. Building on the introductory context, this chapter introduces Minimally Working Programs (MWP) as an instructional tool for teaching computing in physics. It presents productive use cases contextualized within a research-based look at how novices engage with computational models, and it explains why MWPs are valuable across courses, preparing the reader for a discussion of broader resource development.

Chapter 3: M. Lopez del Puerto, L. Engelhardt, and K. Roos; *Resources for Integrating Computation into Physics Courses: Exercise Sets, Workshops, and a Supportive Community*. Expanding from specific tools to community resources, this chapter presents a suite of resources for educators integrating computing into their courses. It highlights classroom activities that are well-organized, open source, and freely available, and describes professional development opportunities within this community, leading into examples of course-level implementations.

Chapter 4: W.B. Lane and T. Galanti; *Computing in General Education Physics*. Transitioning from resources to classroom practice, this chapter introduces a *Physics*

of Music course at the University of North Florida, which integrates computing for general education students. The authors discuss the importance of computing for these students, the learning goals, and strategies for computational instruction, setting up the exploration of early undergraduate research preparation.

Chapter 5: C. Christenson, S. Chamberlin, K. McMichael, and M. McCracken; *Introducing Computational Techniques Early to Train Students for Research*. Continuing the focus on course-level integration, this chapter documents a computing course at Washington and Jefferson College designed to introduce students to computational techniques early in their undergraduate careers. The chapter details course design, learning goals, and the successes and challenges encountered, providing a bridge to curriculum-wide laboratory integration.

Chapter 6: B. Zwickl, C. Lusignan, and A. McGowan; *Instructional Goals and Decisions When Integrating Computation into Physics Labs*. While many integrations have focused on lecture courses, this chapter shifts to the laboratory setting. It discusses the instructional goals and decisions made when integrating computation into physics lab courses, providing a foundation for understanding how computing can enhance experimental learning experiences.

Chapter 7: B. Zwickl, C. Lusignan, and A. McGowan; *Case Studies of Integrating Computation into Physics Lab Courses*. Building directly on the previous chapter, this chapter presents case studies of how computation has been integrated into physics lab courses. It highlights specific examples and outcomes, illustrating the practical applications of computing in experimental physics education.

Chapter 8: R.C. Mastroleo and B. Lunk; *Departmental Emphasis on Computation for the Physics Major: Successes and Failures of a Multi-Tier Curricular Redesign*. Shifting to a broader curricular perspective, this chapter presents a departmental effort to integrate computing into the physics major at Texas State University – San Marcos. It describes how the department designed a curriculum to emphasize computing, the successes and challenges faced, and the lessons learned from the design and implementation process.

Chapter 9: D. O’Neil; *Integration of Computation Across the Curriculum at Bridgewater College*. Extending the theme of curricular integration, this chapter presents how Bridgewater College, a liberal arts school in Virginia, integrated computing throughout its curriculum. The chapter offers lessons for similarly sized and resourced programs and transitions to examples of integration across educational levels.

Chapter 10: M. Hamilton; *Integrating Computation in Physics: From High School Internships to Graduate Studies at Marshall University*. As a final example of curricular integration, this chapter demonstrates how Marshall University faculty have integrated computing across the program, with course offerings for high school through graduate students. The chapter presents examples of student work from each course, setting up a shift to research on teaching and learning.

Chapter 11: K. Svennson, J. Haraldsson, and U. Erickson; *Teachers' Approach to Computation in Physics Classrooms Analyzed Using Social Semiotics and Variation Theory of Learning*. Moving to a more research-focused perspective, this chapter presents a project with high school teachers in Sweden, using social semiotics to understand how teachers and students make meaning of computing tasks. It discusses implications for understanding how learners read program statements and code, leading into innovative instructional approaches.

Chapter 12: T. Atherton, A. Phillips, B. Gravel, and E. Gouvea; *Agency and Making in Computational Learning Environments*. As the complexity of learning computing and making sense of computational tasks becomes clearer, this chapter introduces how agency and the concept of making can be leveraged in computational learning environments. It challenges us to consider agentic learning goals for physics, discusses how these concepts can enhance student engagement and understanding, and provides examples from their computational physics course at Tufts University for how students can take ownership of their learning.

Chapter 13: T. Odden; *Teaching Scientific Writing Using Computational Essays*. The final contributed chapter explores how computational essays can be used to teach scientific writing. It invites us to reflect on how our goals for teaching computing can extend beyond programming skills to encompass broader scientific communication. This chapter presents the concept of computational essays, which combine narrative and code to communicate scientific ideas, and discusses how they can enhance students' understanding of scientific communication and their ability to express complex ideas clearly.

Chapter 14: M.D. Caballero; *Where Do We Go from Here? Next Steps and Considerations*. The final chapter reflects on the lessons learned across the book and considers next steps for integrating computing in physics education, drawing together the themes and insights from all preceding chapters.

5. HOW TO USE THIS BOOK

You are encouraged to explore the chapters that resonate most with your interests and teaching context; each offers distinct insights and practical strategies that can inform your own integration efforts. Collectively, these contributions reflect the diversity, creativity, and dedication of a vibrant community committed to advancing physics education through computing.

Thank you for engaging with this volume. We hope it serves as both a resource and an inspiration as you consider new possibilities for your courses and programs. Most importantly, we invite you to connect with the broader community of educators, share your experiences, and contribute to the ongoing evolution of computing in physics education.

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